

JSS 1-3

INGREDIENT LAB ACTIVITY SYSTEM

IL-301 to IL-400

HYPOTHESIZE -> TEST -> QUANTIFY -> ANALYZE -> EVALUATE

Purpose	Build food and ingredient literacy by connecting plant structure, observation, measurement, post-harvest change, quality and product-development thinking.
Ingredient set	Hibiscus, ginger, turmeric, pineapple, citrus, mint, moringa, watermelon, tomato and maize.
Core rule	The Ingredient Lab is an educational investigation environment. It does not require tasting, medical claims or uncontrolled food processing.
Connections	Link each activity to GrowMeal vocabulary, journals, practical cards, Farm-to-Food activities, missions and assessment.

INGREDIENT LAB - HIBISCUS | PLANT PART: CALYX

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-301 - HIBISCUS - IDENTITY & PLANT PART

Plant part	Calyx
Learning objective	Test a hypothesis about identity & plant part in hibiscus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-302 - HIBISCUS - EXTERNAL STRUCTURE

Plant part	Calyx
Learning objective	Test a hypothesis about external structure in hibiscus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-303 - HIBISCUS - INTERNAL STRUCTURE

Plant part	Calyx
Learning objective	Test a hypothesis about internal structure in hibiscus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-304 - HIBISCUS - COLOUR & TEXTURE

Plant part	Calyx
Learning objective	Test a hypothesis about colour & texture in hibiscus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-305 - HIBISCUS - MASS & SIZE

Plant part	Calyx
Learning objective	Test a hypothesis about mass & size in hibiscus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-306 - HIBISCUS - WATER/CHANGE OBSERVATION

Plant part	Calyx
Learning objective	Test a hypothesis about water/change observation in hibiscus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-307 - HIBISCUS - SORTING & QUALITY

Plant part	Calyx
Learning objective	Test a hypothesis about sorting & quality in hibiscus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-308 - HIBISCUS - POST-HARVEST CHANGE

Plant part	Calyx
Learning objective	Test a hypothesis about post-harvest change in hibiscus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-309 - HIBISCUS - INGREDIENT-TO-FOOD PATHWAY

Plant part	Calyx
Learning objective	Test a hypothesis about ingredient-to-food pathway in hibiscus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-310 - HIBISCUS - INVESTIGATION CHALLENGE

Plant part	Calyx
Learning objective	Test a hypothesis about investigation challenge in hibiscus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - GINGER | PLANT PART: RHIZOME

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-311 - GINGER - IDENTITY & PLANT PART

Plant part	Rhizome
Learning objective	Test a hypothesis about identity & plant part in ginger, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-312 - GINGER - EXTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Test a hypothesis about external structure in ginger, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
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Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-313 - GINGER - INTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Test a hypothesis about internal structure in ginger, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
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Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-314 - GINGER - COLOUR & TEXTURE

Plant part	Rhizome
Learning objective	Test a hypothesis about colour & texture in ginger, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
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Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-315 - GINGER - MASS & SIZE

Plant part	Rhizome
Learning objective	Test a hypothesis about mass & size in ginger, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
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Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-316 - GINGER - WATER/CHANGE OBSERVATION

Plant part	Rhizome
Learning objective	Test a hypothesis about water/change observation in ginger, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
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Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-317 - GINGER - SORTING & QUALITY

Plant part	Rhizome
Learning objective	Test a hypothesis about sorting & quality in ginger, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-318 - GINGER - POST-HARVEST CHANGE

Plant part	Rhizome
Learning objective	Test a hypothesis about post-harvest change in ginger, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-319 - GINGER - INGREDIENT-TO-FOOD PATHWAY

Plant part	Rhizome
Learning objective	Test a hypothesis about ingredient-to-food pathway in ginger, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
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Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-320 - GINGER - INVESTIGATION CHALLENGE

Plant part	Rhizome
Learning objective	Test a hypothesis about investigation challenge in ginger, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
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Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - TURMERIC | PLANT PART: RHIZOME

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-321 - TURMERIC - IDENTITY & PLANT PART

Plant part	Rhizome
Learning objective	Test a hypothesis about identity & plant part in turmeric, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-322 - TURMERIC - EXTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Test a hypothesis about external structure in turmeric, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-323 - TURMERIC - INTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Test a hypothesis about internal structure in turmeric, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-324 - TURMERIC - COLOUR & TEXTURE

Plant part	Rhizome
Learning objective	Test a hypothesis about colour & texture in turmeric, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-325 - TURMERIC - MASS & SIZE

Plant part	Rhizome
Learning objective	Test a hypothesis about mass & size in turmeric, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-326 - TURMERIC - WATER/CHANGE OBSERVATION

Plant part	Rhizome
Learning objective	Test a hypothesis about water/change observation in turmeric, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-327 - TURMERIC - SORTING & QUALITY

Plant part	Rhizome
Learning objective	Test a hypothesis about sorting & quality in turmeric, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-328 - TURMERIC - POST-HARVEST CHANGE

Plant part	Rhizome
Learning objective	Test a hypothesis about post-harvest change in turmeric, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-329 - TURMERIC - INGREDIENT-TO-FOOD PATHWAY

Plant part	Rhizome
Learning objective	Test a hypothesis about ingredient-to-food pathway in turmeric, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-330 - TURMERIC - INVESTIGATION CHALLENGE

Plant part	Rhizome
Learning objective	Test a hypothesis about investigation challenge in turmeric, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - PINEAPPLE | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-331 - PINEAPPLE - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Test a hypothesis about identity & plant part in pineapple, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-332 - PINEAPPLE - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Test a hypothesis about external structure in pineapple, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-333 - PINEAPPLE - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Test a hypothesis about internal structure in pineapple, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-334 - PINEAPPLE - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Test a hypothesis about colour & texture in pineapple, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-335 - PINEAPPLE - MASS & SIZE

Plant part	Fruit
Learning objective	Test a hypothesis about mass & size in pineapple, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-336 - PINEAPPLE - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Test a hypothesis about water/change observation in pineapple, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-337 - PINEAPPLE - SORTING & QUALITY

Plant part	Fruit
Learning objective	Test a hypothesis about sorting & quality in pineapple, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-338 - PINEAPPLE - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Test a hypothesis about post-harvest change in pineapple, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-339 - PINEAPPLE - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Test a hypothesis about ingredient-to-food pathway in pineapple, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-340 - PINEAPPLE - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Test a hypothesis about investigation challenge in pineapple, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - CITRUS | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-341 - CITRUS - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Test a hypothesis about identity & plant part in citrus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-342 - CITRUS - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Test a hypothesis about external structure in citrus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-343 - CITRUS - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Test a hypothesis about internal structure in citrus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-344 - CITRUS - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Test a hypothesis about colour & texture in citrus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-345 - CITRUS - MASS & SIZE

Plant part	Fruit
Learning objective	Test a hypothesis about mass & size in citrus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-346 - CITRUS - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Test a hypothesis about water/change observation in citrus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-347 - CITRUS - SORTING & QUALITY

Plant part	Fruit
Learning objective	Test a hypothesis about sorting & quality in citrus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-348 - CITRUS - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Test a hypothesis about post-harvest change in citrus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-349 - CITRUS - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Test a hypothesis about ingredient-to-food pathway in citrus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-350 - CITRUS - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Test a hypothesis about investigation challenge in citrus, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - MINT | PLANT PART: LEAF

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-351 - MINT - IDENTITY & PLANT PART

Plant part	Leaf
Learning objective	Test a hypothesis about identity & plant part in mint, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-352 - MINT - EXTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Test a hypothesis about external structure in mint, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-353 - MINT - INTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Test a hypothesis about internal structure in mint, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-354 - MINT - COLOUR & TEXTURE

Plant part	Leaf
Learning objective	Test a hypothesis about colour & texture in mint, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-355 - MINT - MASS & SIZE

Plant part	Leaf
Learning objective	Test a hypothesis about mass & size in mint, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-356 - MINT - WATER/CHANGE OBSERVATION

Plant part	Leaf
Learning objective	Test a hypothesis about water/change observation in mint, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-357 - MINT - SORTING & QUALITY

Plant part	Leaf
Learning objective	Test a hypothesis about sorting & quality in mint, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-358 - MINT - POST-HARVEST CHANGE

Plant part	Leaf
Learning objective	Test a hypothesis about post-harvest change in mint, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-359 - MINT - INGREDIENT-TO-FOOD PATHWAY

Plant part	Leaf
Learning objective	Test a hypothesis about ingredient-to-food pathway in mint, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-360 - MINT - INVESTIGATION CHALLENGE

Plant part	Leaf
Learning objective	Test a hypothesis about investigation challenge in mint, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - MORINGA | PLANT PART: LEAF

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-361 - MORINGA - IDENTITY & PLANT PART

Plant part	Leaf
Learning objective	Test a hypothesis about identity & plant part in moringa, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-362 - MORINGA - EXTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Test a hypothesis about external structure in moringa, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-363 - MORINGA - INTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Test a hypothesis about internal structure in moringa, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-364 - MORINGA - COLOUR & TEXTURE

Plant part	Leaf
Learning objective	Test a hypothesis about colour & texture in moringa, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-365 - MORINGA - MASS & SIZE

Plant part	Leaf
Learning objective	Test a hypothesis about mass & size in moringa, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-366 - MORINGA - WATER/CHANGE OBSERVATION

Plant part	Leaf
Learning objective	Test a hypothesis about water/change observation in moringa, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-367 - MORINGA - SORTING & QUALITY

Plant part	Leaf
Learning objective	Test a hypothesis about sorting & quality in moringa, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-368 - MORINGA - POST-HARVEST CHANGE

Plant part	Leaf
Learning objective	Test a hypothesis about post-harvest change in moringa, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-369 - MORINGA - INGREDIENT-TO-FOOD PATHWAY

Plant part	Leaf
Learning objective	Test a hypothesis about ingredient-to-food pathway in moringa, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-370 - MORINGA - INVESTIGATION CHALLENGE

Plant part	Leaf
Learning objective	Test a hypothesis about investigation challenge in moringa, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - WATERMELON | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-371 - WATERMELON - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Test a hypothesis about identity & plant part in watermelon, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-372 - WATERMELON - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Test a hypothesis about external structure in watermelon, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-373 - WATERMELON - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Test a hypothesis about internal structure in watermelon, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-374 - WATERMELON - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Test a hypothesis about colour & texture in watermelon, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-375 - WATERMELON - MASS & SIZE

Plant part	Fruit
Learning objective	Test a hypothesis about mass & size in watermelon, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-376 - WATERMELON - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Test a hypothesis about water/change observation in watermelon, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-377 - WATERMELON - SORTING & QUALITY

Plant part	Fruit
Learning objective	Test a hypothesis about sorting & quality in watermelon, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-378 - WATERMELON - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Test a hypothesis about post-harvest change in watermelon, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-379 - WATERMELON - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Test a hypothesis about ingredient-to-food pathway in watermelon, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-380 - WATERMELON - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Test a hypothesis about investigation challenge in watermelon, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - TOMATO | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-381 - TOMATO - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Test a hypothesis about identity & plant part in tomato, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-382 - TOMATO - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Test a hypothesis about external structure in tomato, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-383 - TOMATO - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Test a hypothesis about internal structure in tomato, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-384 - TOMATO - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Test a hypothesis about colour & texture in tomato, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-385 - TOMATO - MASS & SIZE

Plant part	Fruit
Learning objective	Test a hypothesis about mass & size in tomato, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-386 - TOMATO - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Test a hypothesis about water/change observation in tomato, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-387 - TOMATO - SORTING & QUALITY

Plant part	Fruit
Learning objective	Test a hypothesis about sorting & quality in tomato, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-388 - TOMATO - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Test a hypothesis about post-harvest change in tomato, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-389 - TOMATO - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Test a hypothesis about ingredient-to-food pathway in tomato, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-390 - TOMATO - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Test a hypothesis about investigation challenge in tomato, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - MAIZE | PLANT PART: SEED/GRAIN

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-391 - MAIZE - IDENTITY & PLANT PART

Plant part	Seed/Grain
Learning objective	Test a hypothesis about identity & plant part in maize, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-392 - MAIZE - EXTERNAL STRUCTURE

Plant part	Seed/Grain
Learning objective	Test a hypothesis about external structure in maize, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-393 - MAIZE - INTERNAL STRUCTURE

Plant part	Seed/Grain
Learning objective	Test a hypothesis about internal structure in maize, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-394 - MAIZE - COLOUR & TEXTURE

Plant part	Seed/Grain
Learning objective	Test a hypothesis about colour & texture in maize, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-395 - MAIZE - MASS & SIZE

Plant part	Seed/Grain
Learning objective	Test a hypothesis about mass & size in maize, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-396 - MAIZE - WATER/CHANGE OBSERVATION

Plant part	Seed/Grain
Learning objective	Test a hypothesis about water/change observation in maize, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-397 - MAIZE - SORTING & QUALITY

Plant part	Seed/Grain
Learning objective	Test a hypothesis about sorting & quality in maize, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-398 - MAIZE - POST-HARVEST CHANGE

Plant part	Seed/Grain
Learning objective	Test a hypothesis about post-harvest change in maize, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-399 - MAIZE - INGREDIENT-TO-FOOD PATHWAY

Plant part	Seed/Grain
Learning objective	Test a hypothesis about ingredient-to-food pathway in maize, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-400 - MAIZE - INVESTIGATION CHALLENGE

Plant part	Seed/Grain
Learning objective	Test a hypothesis about investigation challenge in maize, quantify variation or change, and evaluate implications for ingredient quality or handling.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Define hypothesis and variables; label replicates; collect traceable measurements; calculate a summary/effect; graph where useful; identify uncertainty; recommend a handling or quality decision.
Evidence to record	Hypothesis, protocol, replicate data, quantitative analysis, graph, uncertainty and recommendation.
Questions	What is the response variable? How much variation is present? Does the evidence support the hypothesis? What decision follows?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?